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Gas Leakage Detection System

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PROJECT OVERVIEW

- This project is a gas leakage detection system designed to detect the presence of gases like LPG, natural gas, and methane using an MQ5 gas sensor.
- Upon detection of gas beyond a threshold level, an LED indicator turns on to alert the user. The system is simple, cost-effective, and can be integrated into home or industrial safety setups.

COMPONENTS USED:

1. MQ5 Gas Sensor

- Function: Detects gases such as LPG, natural gas, and hydrogen.
- Specifications:
  - Detection Range: 200-10000 ppm (parts per million)
  - Operating Voltage: 5V DC
  - Analog Output Voltage: 0.1V to 4.0V
  - Preheat Time: ~20 seconds
  - Sensitivity: Adjustable via potentiometer (on sensor module)
- Working Principle: It uses a SnO_2 (tin dioxide) sensing layer whose conductivity increases in the presence of gas

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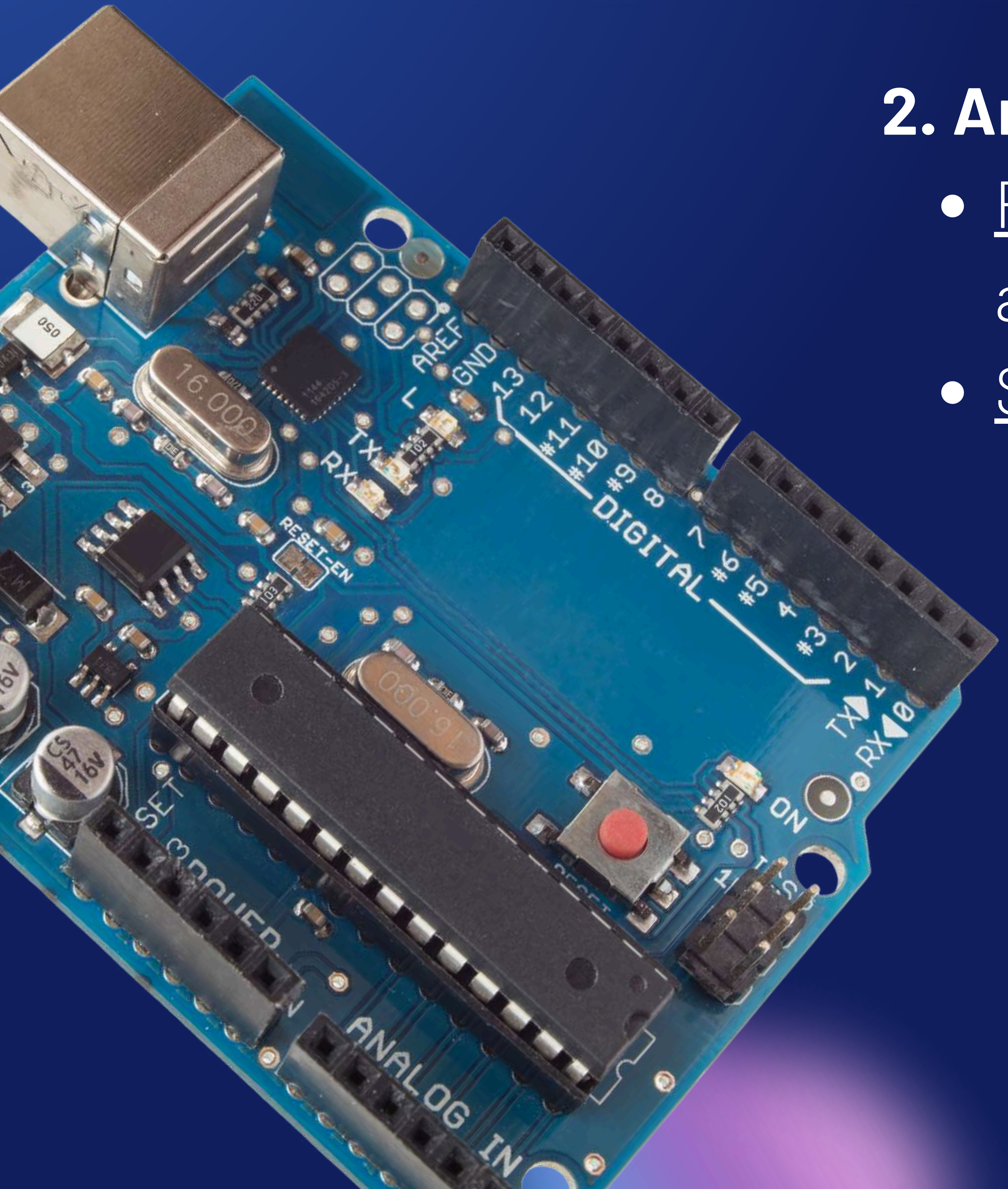
COMPONENTS USED:



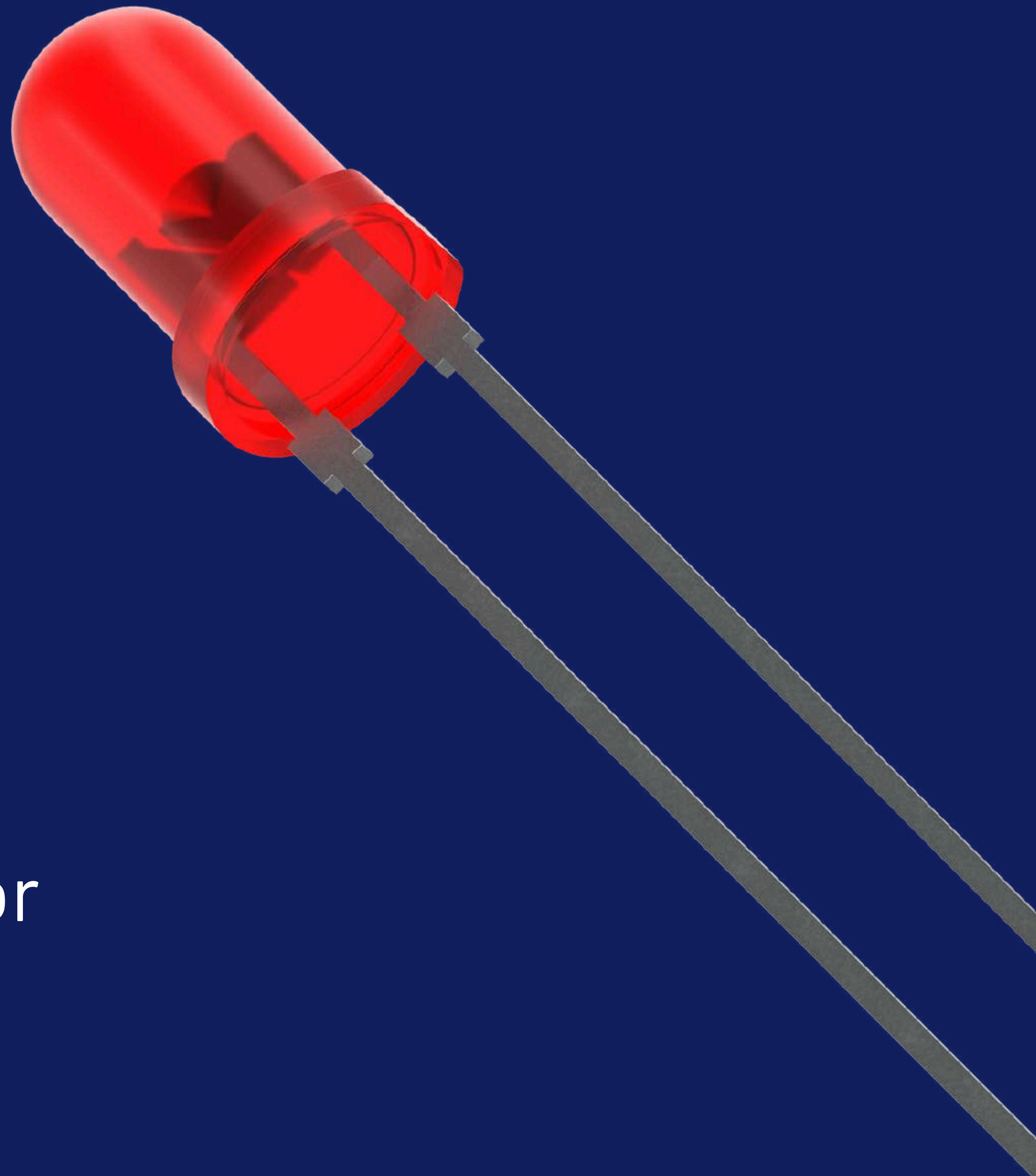
2. Arduino Uno

- Function: Acts as the microcontroller that reads the analog signal from the MQ5 and processes it.
- Specifications:
  - Microcontroller: ATmega328P
  - Operating Voltage: 5V
  - Analog Input Pins: 6 (A0–A5)
  - Clock Speed: 16 MHz
  - Flash Memory: 32 KB
  - Communication Ports: USB, UART, I2C, SPI

COMPONENTS USED:

5. LED (Light Emitting Diode)

- Function: Acts as a visual alert when gas is detected.
- Specifications:
  - Forward Voltage: ~2V
  - Operating Current: 20mA
  - Color: Typically red or any other warning color
  - Connected with Resistor: Typically 220Ω resistor to prevent damage



COMPONENTS USED:

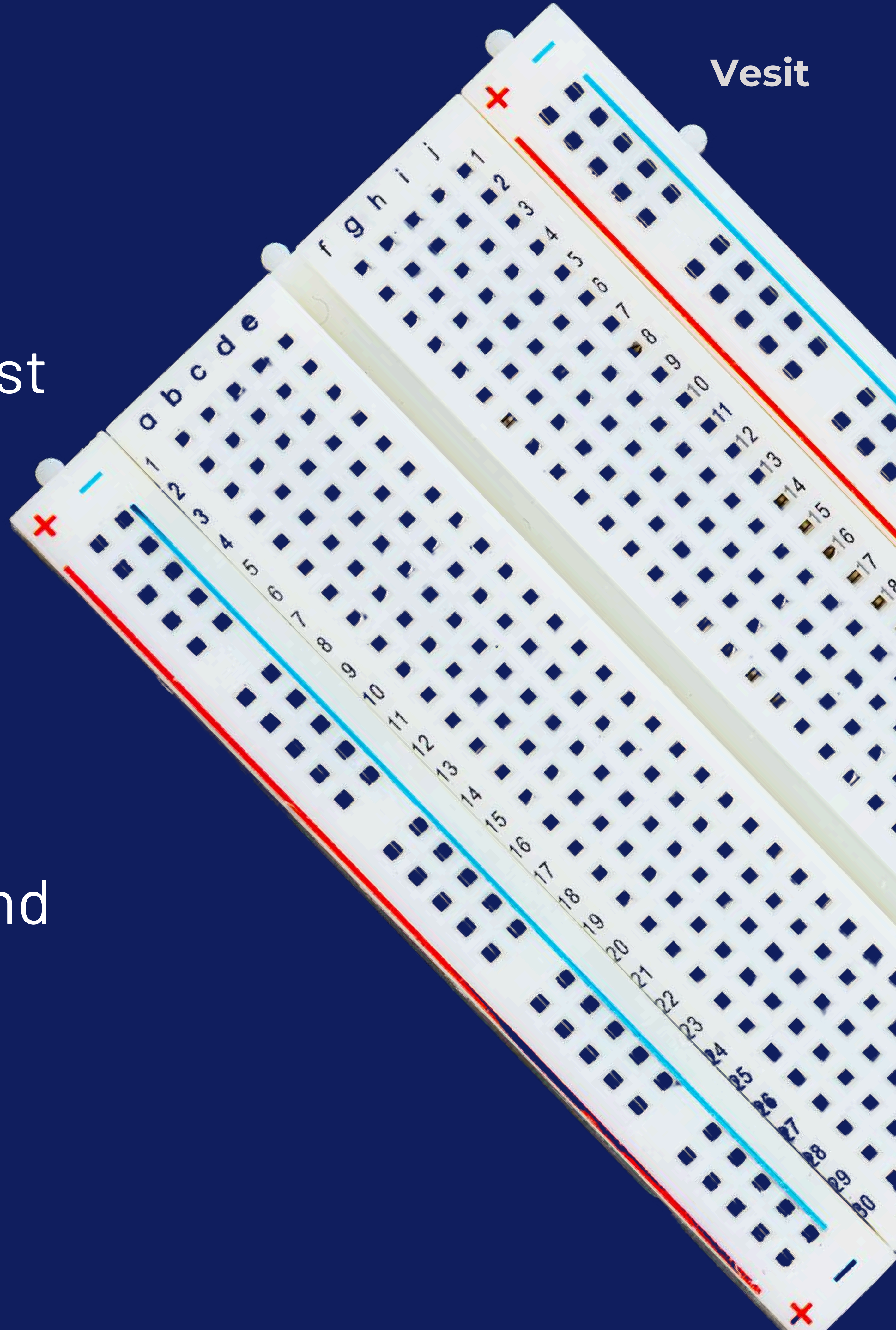
4. Jumper Wires

- Function: Used to connect components on the breadboard with the Arduino and each other.
- Types:
  - Male-to-Male
  - Female-to-Male
- Standard Lengths: 10–20 cm

COMPONENTS USED:

3. Breadboard

- Function: A temporary platform to build and test the circuit without soldering.
- Specifications:
  - Type: Full-size or half-size breadboard
  - Rows/Columns: Typically 30 rows and 10 columns
  - Power Rails: For easy connection of VCC and GND



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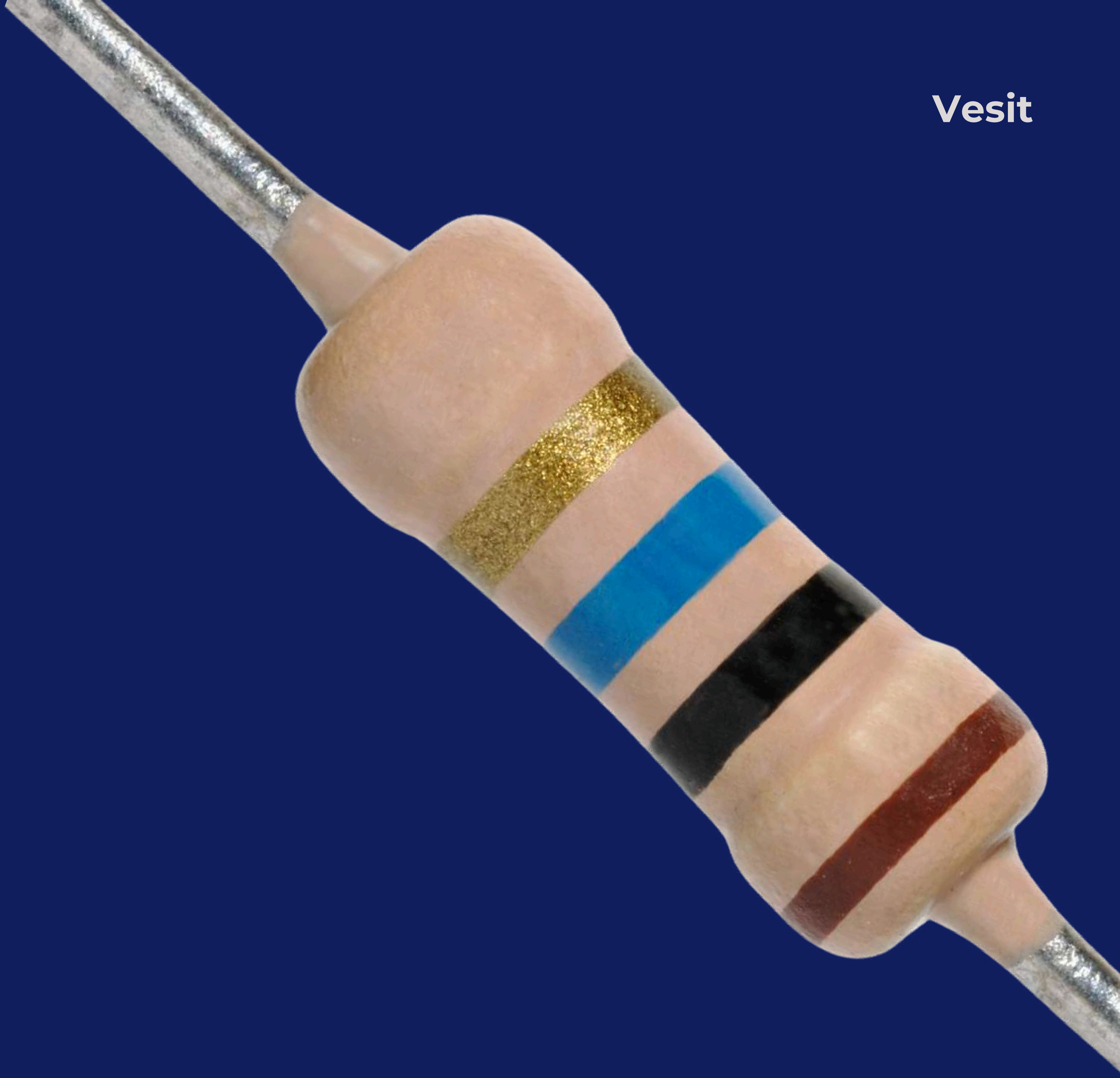
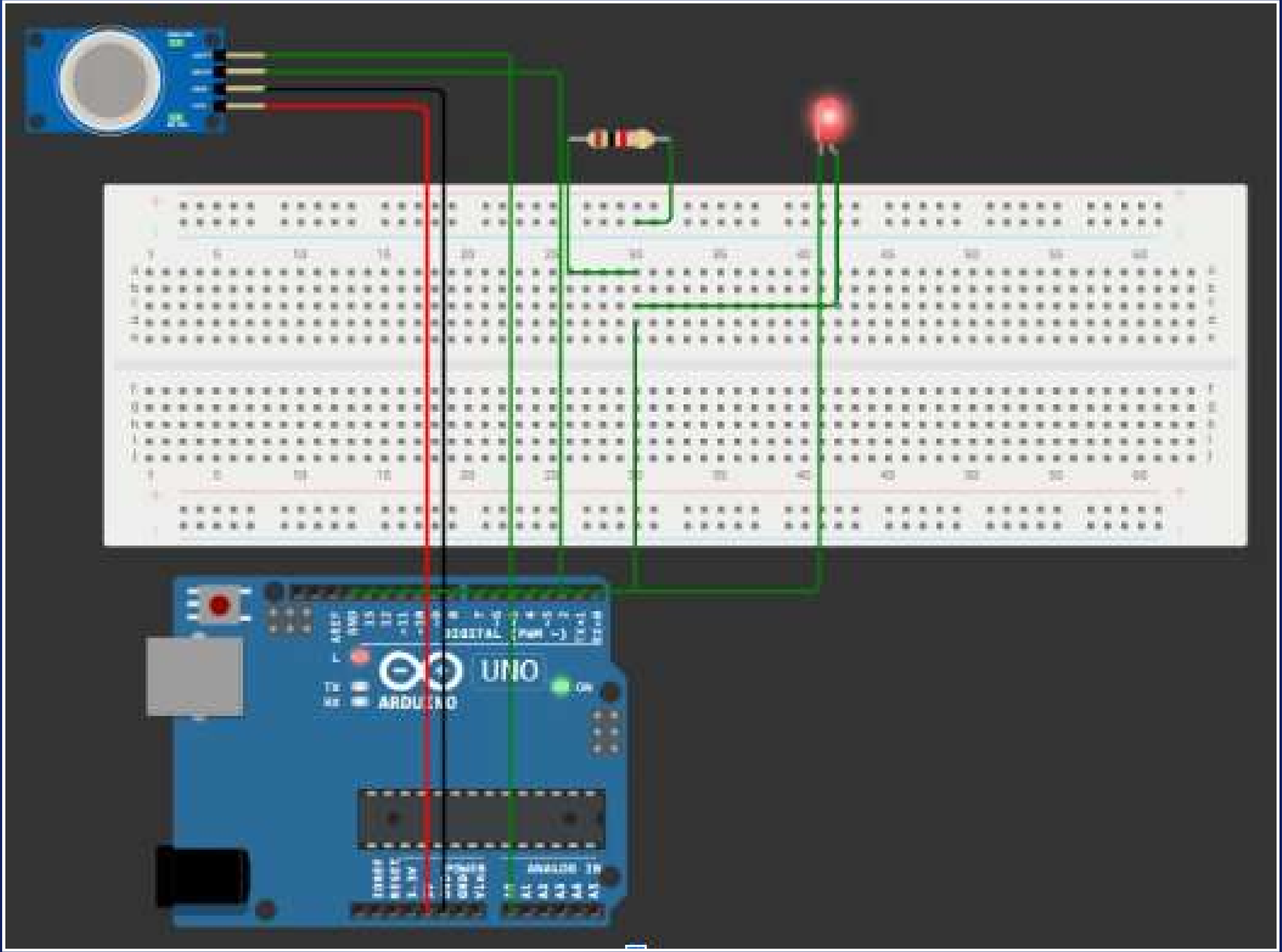
COMPONENTS USED:



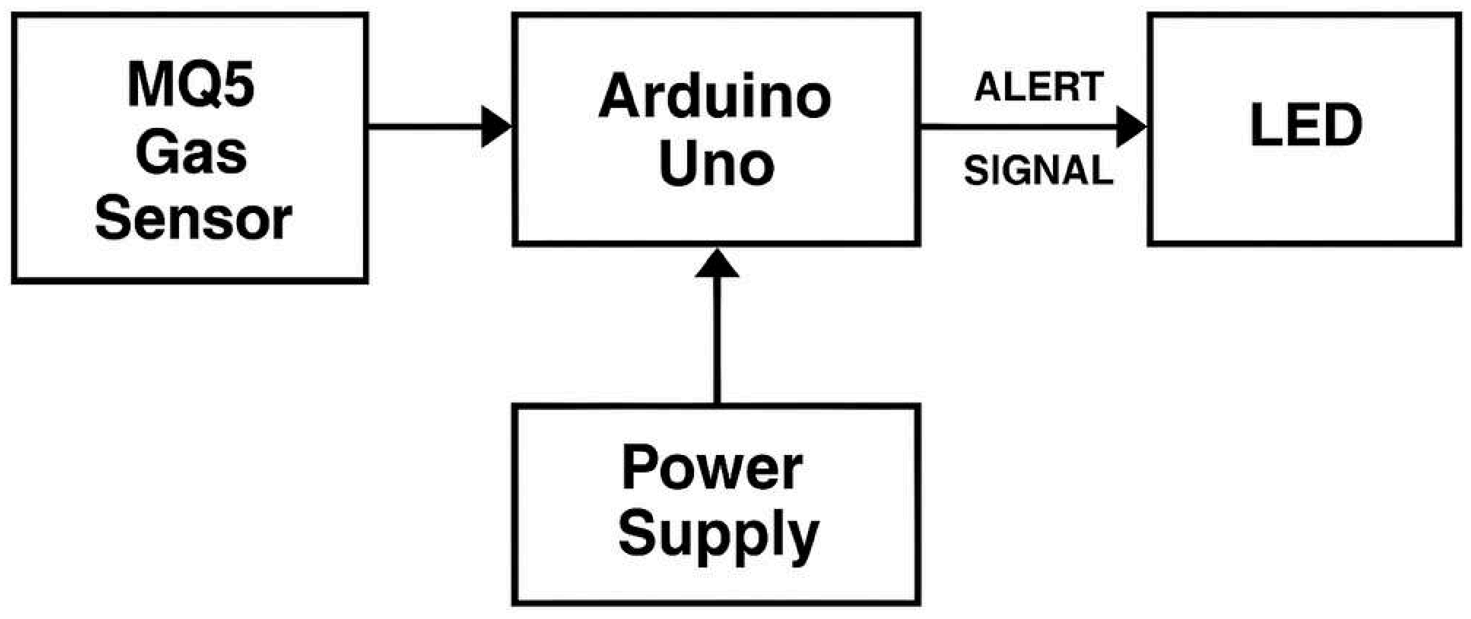
6. Resistor

- Function: Limits the current flowing through the LED.
- Specification: 220Ω to 330Ω

SIMULATION



BLOCK DIAGRAM



Block Diagram

DETAILED WORKFLOW

1. Power Supply Initialization

- When the system is powered on using a 5V DC supply (via USB or external adapter), the Arduino Uno and the MQ5 gas sensor are energized.
- The MQ5 sensor begins its preheat phase, which takes approximately 20–30 seconds to stabilize.
- During this time, its internal heating coil heats up the sensor's SnO_2 layer, preparing it to detect gases effectively.

DETAILED WORKFLOW

2. Sensor Monitoring Begins

- The Arduino Uno reads analog voltage values from the A0 pin connected to the MQ5 sensor's analog output.
- These values range from 0V to 5V, corresponding to a gas concentration range of 200 ppm to 10,000 ppm (parts per million), depending on the type of gas.
- Internally, the Arduino converts this voltage to a digital value between 0 and 1023 (since it has a 10-bit ADC).

DETAILED WORKFLOW

3. Threshold Comparison

- A predefined threshold value (for example, an analog reading of 400, which corresponds to approximately 1.46V) is set in the Arduino code.
- This threshold is based on safe gas concentration limits—approximately 1000 ppm depending on calibration.
- If the analog reading is below 300 (≈1.46V), it means the gas level is within safe limits.

DETAILED WORKFLOW

4. Leakage Detection Decision

- The Arduino compares the sensor's output against the threshold:
  - If < 400 ($\approx 1.46V$ or < 1000 ppm) $\rightarrow$ Environment is safe.
  - If ≥ 400 ($\geq 1.46V$ or ≥ 1000 ppm) $\rightarrow$ Possible gas leakage is detected.

DETAILED WORKFLOW

5. Alert Activation

- Upon detecting a gas concentration above the threshold:
  - The Arduino outputs HIGH (5V) on a digital pin (e.g., D13).
  - An LED connected via a 220Ω resistor lights up as a visual warning.
  - Optionally, a buzzer or relay can be triggered simultaneously.

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DETAILED WORKFLOW

6. Continuous Monitoring

- The system continues to monitor gas concentration every few milliseconds using the loop() function in Arduino.
- If the gas level drops below the threshold:
  - The LED is turned OFF, indicating safety has been restored.
- The system is self-resetting and continues real-time detection without manual resets.

DETAILED WORKFLOW

7. System Safety & Expansion Possibilities

- The system runs continuously on 5V, drawing minimal current (typically under 200 mA).
- Expansion ideas:
  - GSM Module (e.g., SIM800L): Sends SMS alerts when gas is detected.
  - Relay Module: Automatically switches on exhaust fans or shuts gas valves.
  - OLED/LCD Display: Shows real-time gas concentration in ppm.
  - IoT integration: Send alerts to smartphones via Wi-Fi or cloud.

SUMMARY

1. System powered at 5V DC, sensor warms up (20–30 sec)
2. Arduino reads gas level (0–5V analog, 200–10,000 ppm)
3. Compares to threshold (e.g., 300 analog units ≈ 1.46V ≈ 1000 ppm)
4. If below threshold → No alert
5. If above threshold → LED ON (5V output, 20mA, 220Ω resistor)
6. Continuously checks sensor value and updates alert status
7. Fully automatic, real-time, and expandable system